

Nick DiSanto

<https://nickdisanto.github.io>

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Education	Vanderbilt University PhD in Computer Science GPA: 4.0/4.0.	Jan. 2025 – Present
	California Baptist University B.S. in Computer Science, <i>summa cum laude</i> GPA: 4.0/4.0. Ranked #1 in graduating class.	Aug. 2019 – Apr. 2023
Experience	Vanderbilt Medical Image Computing Lab <i>Graduate Student Researcher</i> • Using deep learning for image restoration and interpolation, particularly in the medical domain • Building registration and inpainting models for 3D OCT volumes	Jan. 2025 – Present
	GE Vernova <i>Software Engineer</i> • Led the wildfire forecasting initiative, including the end-to-end training, visualization, and productization of ML pipelines with power grid applications • Presented results of wildfire risk forecasting analytic to the C-Suite, helping to secure significant funding for further integration • Honored as a 2024 GE Vernova Changemaker for co-leading a company-wide initiative to integrate GenAI into GEV Electrification Software products	Jan. 2024 – Jan. 2025
	CalBaptist ML & NLP Lab <i>Lead Undergraduate Researcher</i> • Built representation models to find empirical patterns in unstructured online data, challenging the necessity of complex algorithms for natural language pattern recognition • Proposed a novel data preprocessing approach that introduces noise and helps medical imaging models better perceive their environments, increasing accessibility for low-income communities • Produced analyses and applications for three papers, leading the team from inception to execution	Jan. 2022 – Dec. 2023
	Sirch <i>NLP Consultant</i> • Built models that perform Query Auto-Completion (QAC), information retrieval, and NLU • Scaled product to efficiently handle increasing foot traffic while maintaining a personalized UI	Sep. 2023 – Dec. 2023
	General Electric <i>Software Engineering Intern</i> • Led a database optimization initiative that saves thousands of dollars per year • Developed and optimized microservices for large-scale data ingestion applications	June 2022 – Aug. 2023
	Keysight Technologies <i>Software Engineering Intern</i> • Created a web platform and built applications to enable clients to customize their products • Performed data analysis to help eliminate 40% of product-option configurations	June 2021 – Aug. 2021

Publications & Preprints	<i>LOTUS: Latent Outpainting Diffusion Model for Three-Dimensional Ultrasound Stitching.</i> X Yao, R Yu, N DiSanto , E Aghdam, K Oguine, D Lu, A Lou, J Wang, D Hu, G Arenas, B Oguz, A Pouch, N Schwartz, B Byram, I Oguz. <i>Medical Imaging with Deep Learning (MIDL)</i> , 2025. [code]	
	<i>Spatial Analysis of Social Media's Proxies for Human Emotion and Cognition.</i> A Corso, N DiSanto , N Corso, E Lee. <i>International Conference on Information</i> , 2024.	
	<i>Transcending the Attention Paradigm: Representation Learning from Geospatial Social Media Data.</i> N DiSanto , A Corso, B Sanders, G Harding. <i>arXiv preprint arXiv:2310.05378</i> , 2023. [pdf] [code]	
	<i>Leveraging Contextual Data Augmentation for Generalizable Melanoma Detection.</i> N DiSanto , G Harding, E Martinez, B Sanders. <i>arXiv preprint arXiv:2212.05116</i> , 2022. [pdf] [code]	
	<i>Beyond Interpretable Benchmarks: Contextual Learning through Cognitive and Multimodal Perception.</i> N DiSanto . <i>arXiv preprint arXiv:2304.00002</i> , 2022. [pdf]	
Teaching Experience	EGR121: Intro to C++ , <i>Teaching Assistant</i>	SP 2023
	EGR225: Discrete Structures , <i>Teaching Assistant</i>	FA 2022
	CSC312: Algorithms , <i>Tutor</i>	SP/FA 2022
	EGR329: Computer Architecture , <i>Tutor</i>	SP/FA 2022
	PHY201: Physics for Engineers , <i>Teaching Assistant</i>	SP/FA 2021
Service	Reviewer • <i>Decision Support Systems</i>, 2023 • <i>Mathematics</i>, 2022	
Awards & Honors	2024 GE Vernova Changemaker (selected out of 1200 nominees)	Nov. 2024
	CS Outstanding Student Award (ranked #1/40)	Apr. 2023
	Inducted into Alpha Chi Honor Society (top 10%)	Apr. 2022
	Physics Performance & Leadership Award	Dec. 2021
	President's List (every semester of undergraduate study)	FA 2019 – SP 2023
	Trustee Merit Scholarship	FA 2019 – SP 2023
	Engineering Excellence Scholarship	SP 2021 - SP 2023
Leadership & Outreach	Founder and President of CBU Chess Club.	
	Member of CBU's Association for Computing Machinery (ACM) Chapter.	
	Lead trumpet and section leader in 4 competitive CBU musical ensembles.	
	Set up a newly-opened local high school's computer lab.	
	Performed STAR test data analysis for an underresourced elementary school during COVID-19 pandemic.	